TEL AVIV UNIVERSITY אוניברסיטת תל אביב

Approximation Algorithms for Big Data

Lecture 8: Efficient linear algebra & LLMs

Spring 2026

Instructor: Tal Wagner

Administrative Announcements

- Final project guidelines published on Moodle

- New timeline:
  - **July 11:** proposal submission
  - **July 21**: proposal re-submission (if needed)
  - **August 27:** final report submission
  - **Sep 6-30:** final project presentations

- Please read the guidelines carefully
- Remember: projects are required to delve into full details
  - “High-level overview” projects will not be graded well

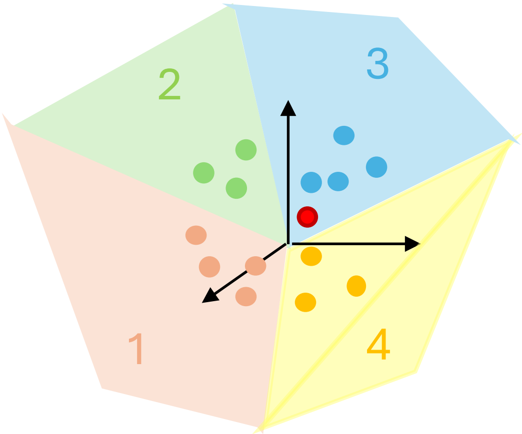
Plan

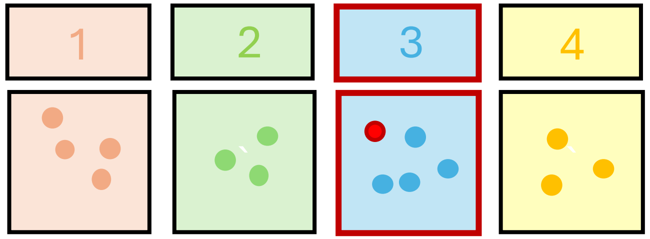
- Recap last time (briefly)

- Today's topic 1: Efficient approximate linear algebra
  - Review linear algebra background
    - Matrix norms; orthogonal projections; SVD; pseudo-inverse
  - Approximate matrix multiplication via sketching
  - Approximate least-squares regression via sketching

- Today's topic 2: Use of sketching techniques in LLMs
  - Informal taste

Recap





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  - Importance sampling in general
  - Importance sampling for KDE
  - Hashing-based estimators (HBE): KDE with relative error via LSH

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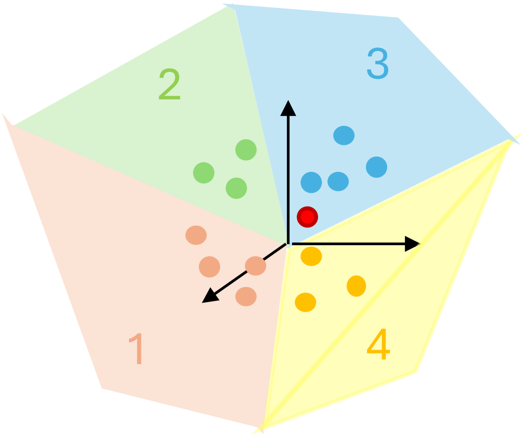

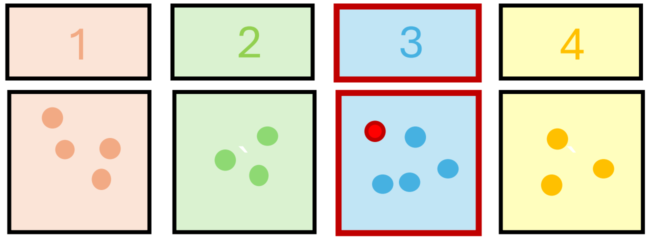

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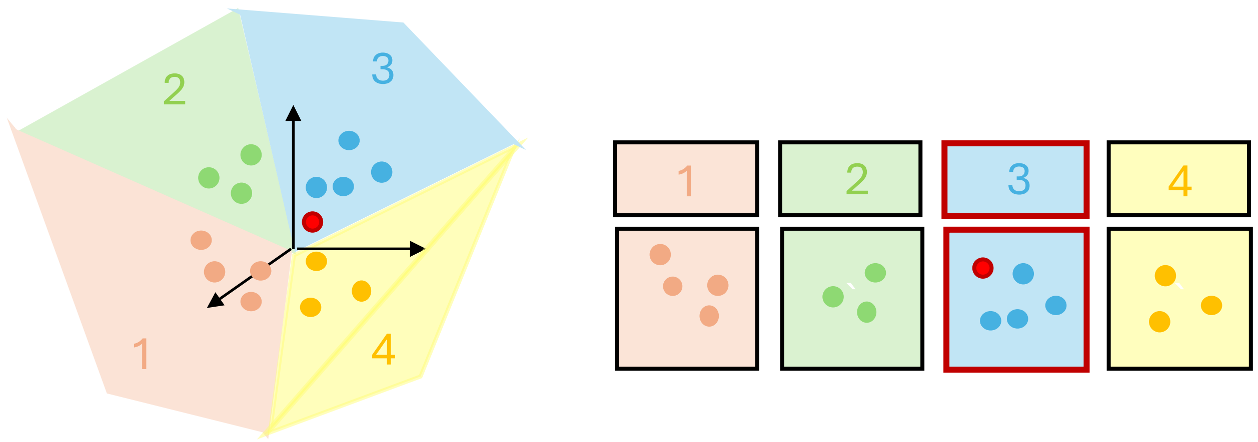

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  - Final project commentary

Fast Approximate Linear Algebra

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- Extremely computationally taxing

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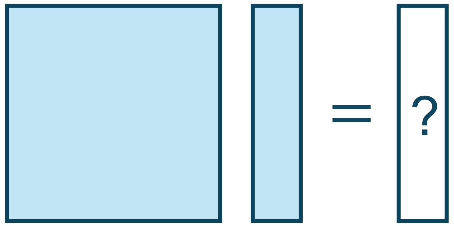
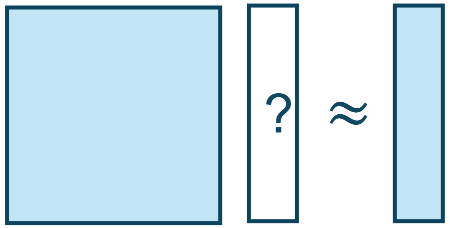
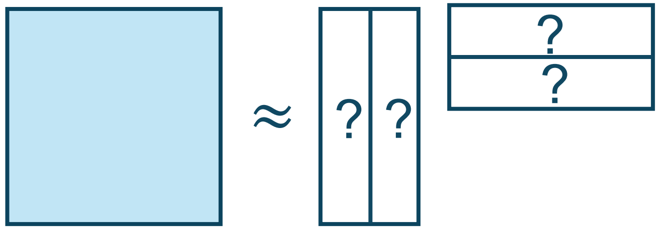


Matrix-vector / matrix-matrix multiplication



Linear regression



Low-rank approximation

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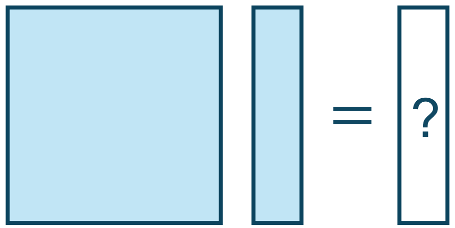
- Why would the techniques we've seen so far be related?



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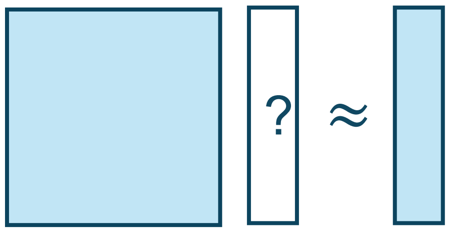

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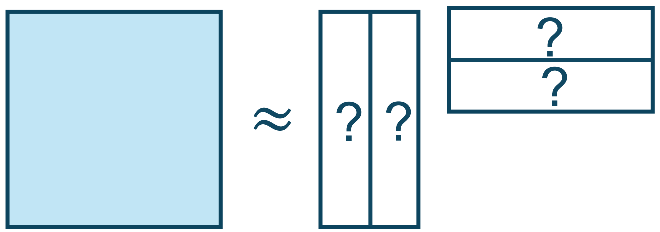

Low-rank approximation

Linear Algebra Background

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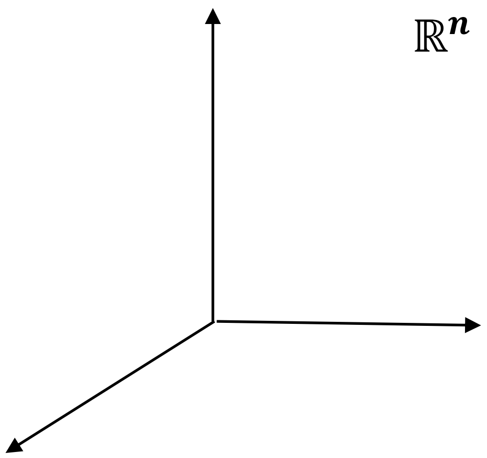
Orthogonal Projections



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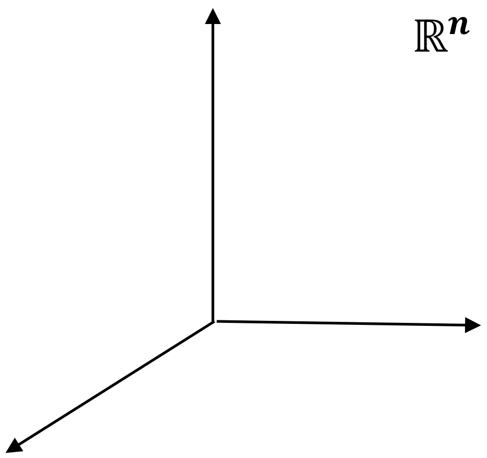

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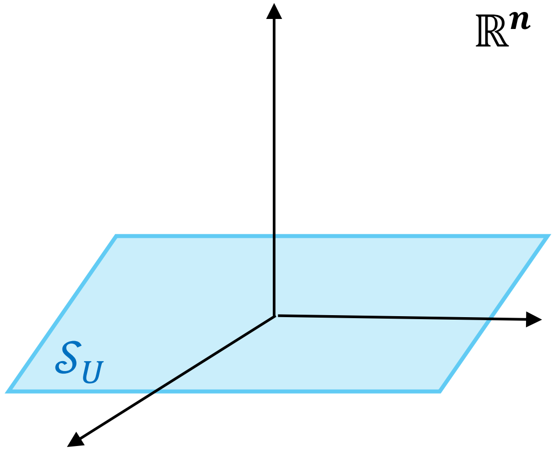

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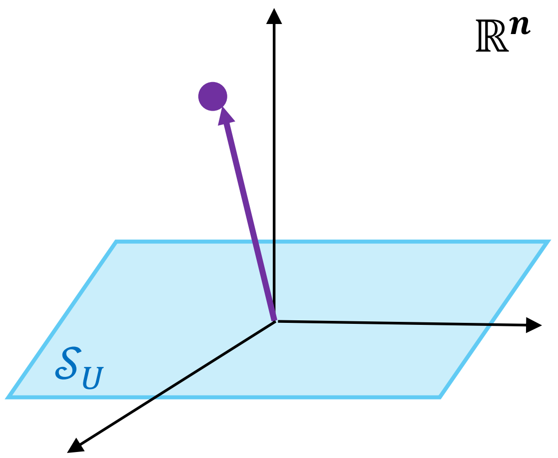

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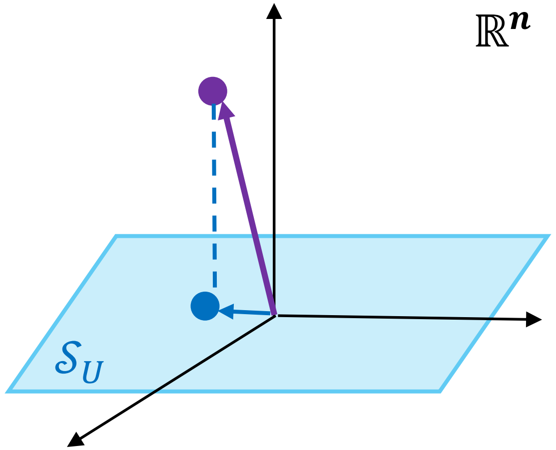

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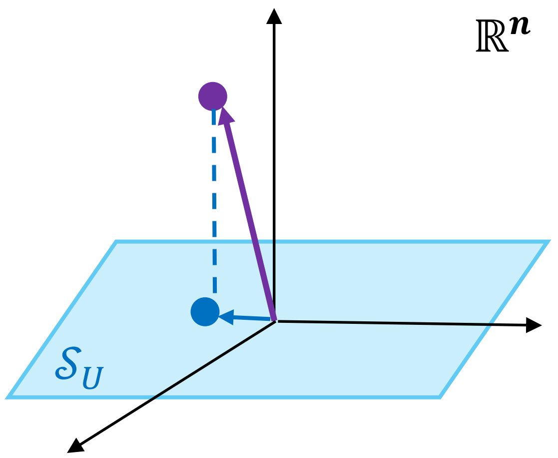

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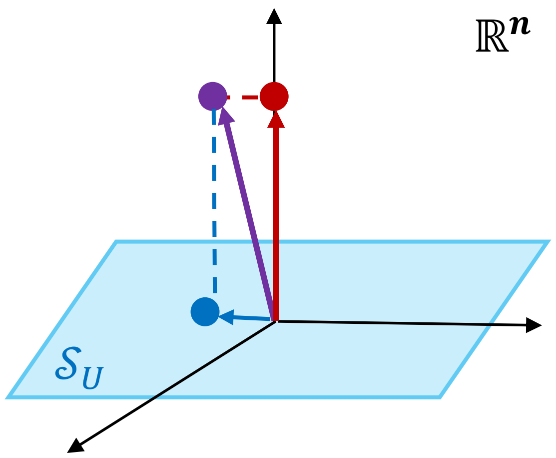

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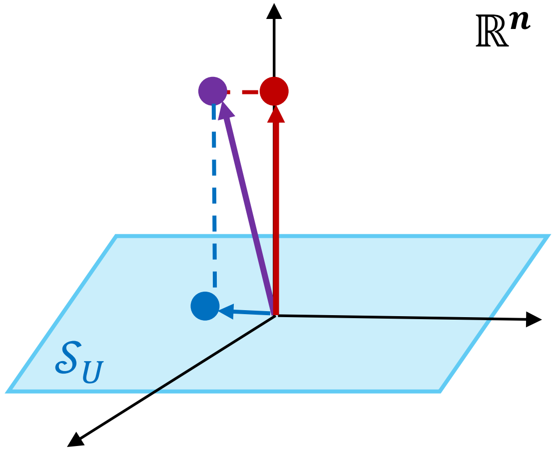

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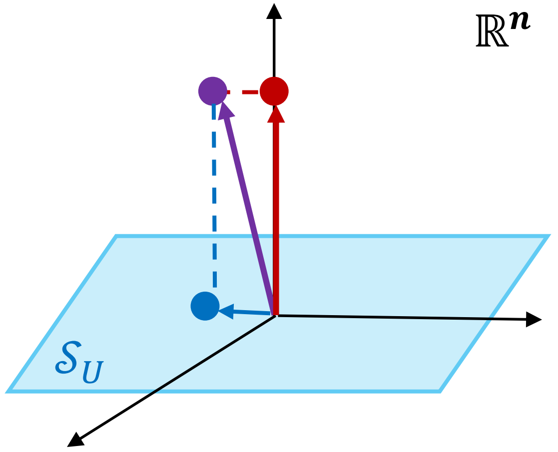

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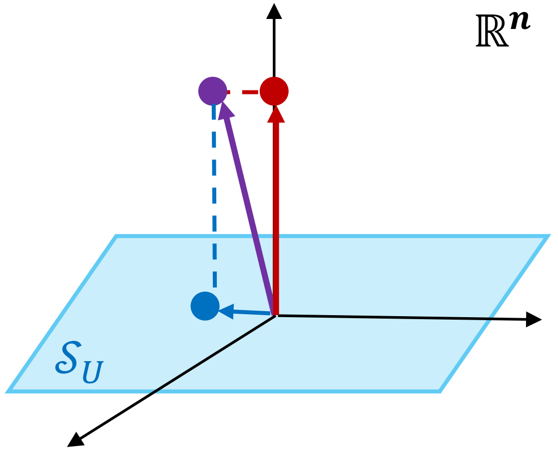

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- The Pythagorean identity: $\|x\|_2^2 = \|UU^T x\|_2^2 + \|(I - UU^T)x\|_2^2$ (proof?)
- Extended to matrices $M \in \mathbb{R}^{n \times m}$: $\|M\|_F^2 = \|UU^T M\|_F^2 + \|(I - UU^T)M\|_F^2$ (proof?)

Singular Value Decomposition (SVD)

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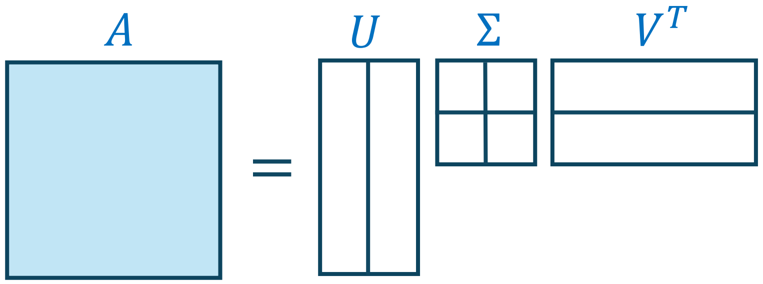

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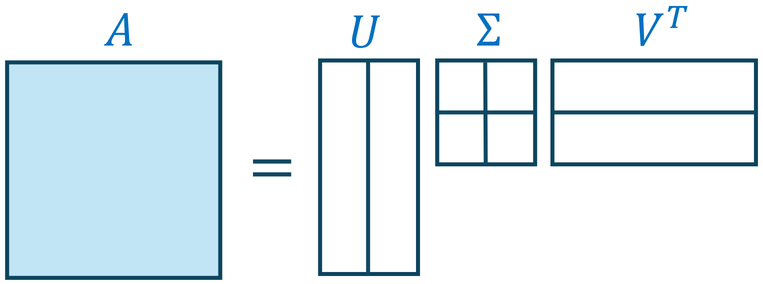

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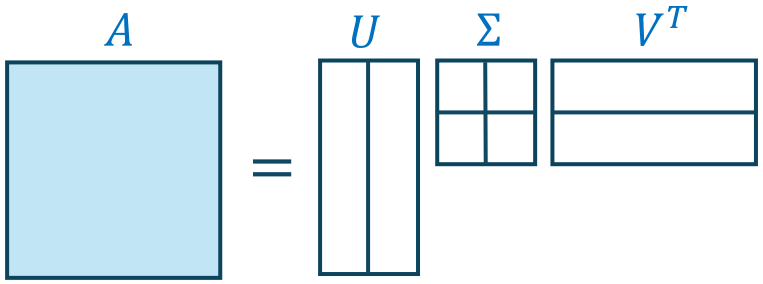

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- Equivalent writing: $A = \sum_{i=1}^{r} \sigma_i \cdot u_i v_i^T$ (A as the sum of r rank-1 matrices)
- The σ_i are the singular values of A
  - The columns of U are called left-singular vectors
  - The columns of V are called right-singular vectors

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- Fact 2: The Frobenius norm is $\boxed{\text{Expression as singular values ???}}$

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  - Use two properties that are easy to verify by direct calculation:
    1. $\|A\|_F = trace(A^T A)$
    2. $trace(BC) = trace(CB)$ for any two matrices B, C (“trace is cyclic”)

Moore-Penrose Pseudo-Inverse

- The Moore-Penrose pseudo-inverse A^+ of a matrix $A \in \mathbb{R}^{n \times d}$ is a generalization of the matrix inverse that exists (uniquely) for every matrix $A \in \mathbb{R}^{n \times d}$
  - Even if A is square with non-full rank
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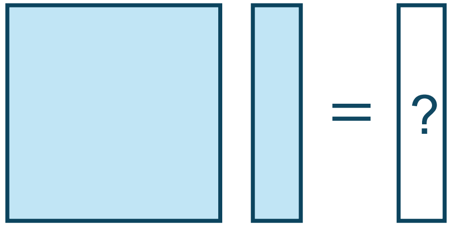
- A^+ is the unique matrix that satisfies all four properties

Approximate Matrix Multiplication



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- Approach – **"Sketch and Solve":**
  - Sample a random sketching matrix $S \in \mathbb{R}^{m\times n}$
  - Sketch: compute SA and SB
  - Solve: return $C = (SA)^T (SB)$ as the approximation for $A^T B$

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  - Sketch: compute SA and SB
  - Solve: return $C = (SA)^T(SB)$ as the approximation for $A^T B$
  - Which random sketching matrix can we use?

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- What is the running time?
- Do you think it can be improved? How?

The JL Moment Property (JLMP)

- <u>Definition</u>: a distribution over random matrices $S \in \mathbb{R}^{m \times n}$ satisfies the (ε, δ)-second moment JL property (2-JLMP) if for every unit vector $x \in \mathbb{R}^n$:

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- Hence: $\mathbb{E}[\|Sx\|_2^2] = \mathbb{E}[\sum_{j=1}^m (Sx)_j^2] = 1$

From JLMP to AMM (1/2)

2-JLMP:

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  - We have the identity: $x^T y = \frac{1}{4}(\|x+y\|_2^2 + \|x-y\|_2^2)$, hence:

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  - $\le \frac{1}{4}\sqrt{\mathbb{E}[(\|S(x+y)\|_2^2 - \|x+y\|_2^2)]^2} + \sqrt{\mathbb{E}[(\|S(x-y)\|_2^2 - \|x-y\|_2^2)]^2}$ Why?
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Corollary:

$$\forall x, y \in \mathbb{R}^n\colon \mathbb{E}[x^T S^T S y - x^T y]^2 \le \varepsilon\sqrt{\delta}\|x\|_2^2\|y\|_2^2$$

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Lemma from previous slide:

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- $= \varepsilon^2\delta\, \|A\|_F^2\|B\|_F^2$
- Hence by Markov: $\Pr[\|(SA)^T(SB) - A^T B\|_F^2 > \varepsilon^2\|A\|_F\|B\|_F^2] < \delta$.

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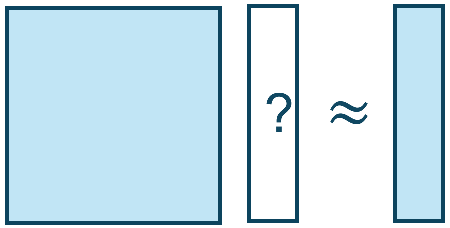
- Hence by Markov: $\Pr[\|(SA)^T(SB) - A^T B\|_F^2 > \varepsilon^2\|A\|_F\|B\|_F^2] < \delta.$
- We proved the AMM via JL theorem – what running time did we get?

Regression



ℓ_2-Regression

- Definition: in the ℓ_2-regression problem,
  - Input: $A \in \mathbb{R}^{n \times d}$ and $b \in \mathbb{R}^d$
  - Output: $x^* = \min_{x \in \mathbb{R}^d} ||Ax - b||_2$ (i.e., solve the linear equation as best as possible)

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- Ideas how to achieve this?

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- Why are these the same?

Approximate Regression via OSE

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- How can we solve ℓ_2-regression with an OSE?
- Sketch-and-solve:
  - Sample S
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- **Hence**: Our next goal is to build OSEs (ideas?)

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- **Corollary:** $S \in \mathbb{R}^{m \times n}$ with iid Gaussians and $m = O\left(\frac{\log(N/\delta)}{\varepsilon^2}\right)$ satisfies (ε, δ, N)-IPJL

OSE via IPJL (1/4)

- Let S be a random matrix satisfying (ε, δ, N)-IPJL
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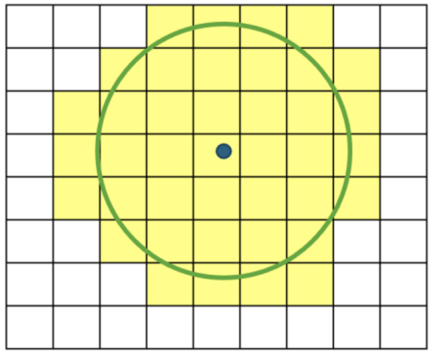

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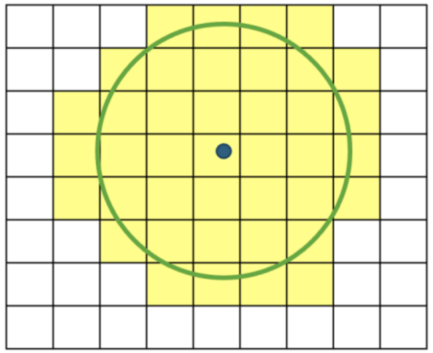
  - We will take $\eta = 1/2$, so $|Z| = O(1)^d$



OSE via IPJL (2/4)

- Recall: we saw that have $S \in \mathbb{R}^{m \times n}$ with $m = O\left(\frac{\log(N/\delta)}{\varepsilon^2}\right)$ satisfies (ε, δ, N)-IPJL
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- Recall: Z is a $\frac{1}{2}$-net of size $O(1)^d$ for the unit sphere in $\mathcal{V}$
- We apply IPJL to every point in Z
- **Thus:**
  - S has $m = O\left(\frac{d \log(1/\delta)}{\varepsilon^2}\right)$ rows
  - For every $z, z' \in Z$ we have:

$$\Pr[\forall\, i, j\colon\ z^T S^T S z' = z^T z' \pm \varepsilon \|z\|_2 \|z'\|_2] \geq 1 - \delta$$

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  - We thus have:

$$x = \sum_{i=1}^{\infty} z_i \text{ such that } \forall i, \|z_i\|_2 = \frac{1}{2^i}$$

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- And, $\sum_{i,j=1}^{\infty} \|z_i\|_2 \left\|z_j\right\|_2 = \sum_{i,j=1}^{\infty} \frac{1}{2^i} \cdot \frac{1}{2^j} = \left(\sum_{i=1}^{\infty} \frac{1}{2^i}\right)^2 = 1.$

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- **We proved**: JL=>IPJL=>OSE, thus: every distribution over matrices that satisfies usual JL yields an OSE with $\tilde{O}(d/\varepsilon^2)$ rows.

Approximate Regression via OSE with JL

- **Corollary**:
  - Let $A \in \mathbb{R}^{n \times d}$, $b \in \mathbb{R}^d$ be a regression problem.
  - Let $m = \tilde{O}(d/\varepsilon^2)$.
  - Let $S \in \mathbb{R}^{m \times n}$ be a random matrix with iid Gaussian entries $\frac{1}{\sqrt{m}} N(0,1)$.
  - Then, the sketch-and-solve solution $\tilde{x} = \min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2$ satisfies the approximate regression guarantee:

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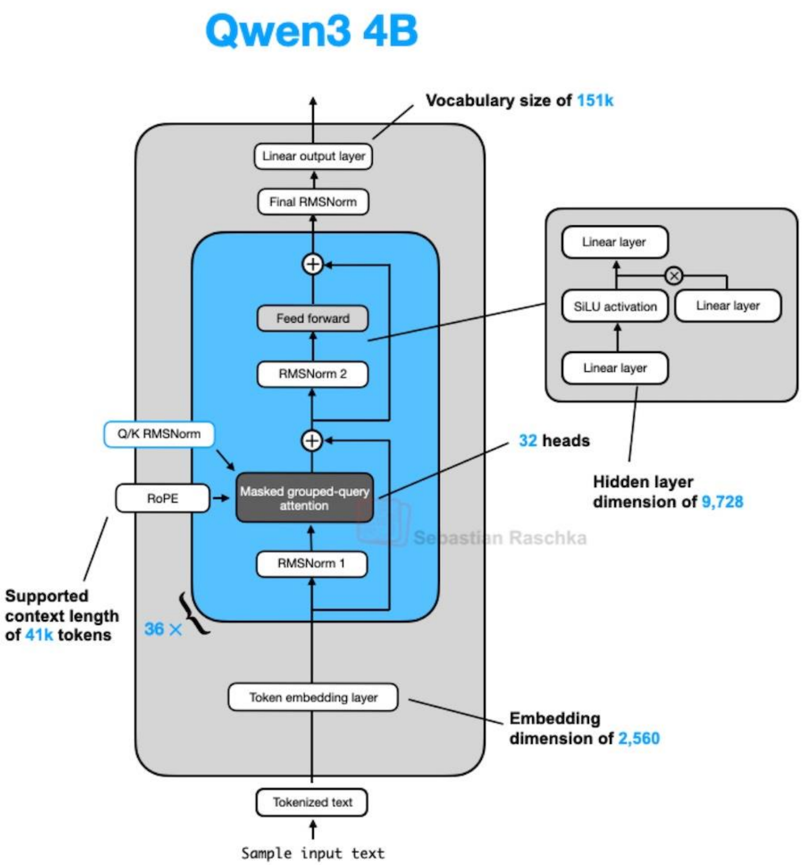
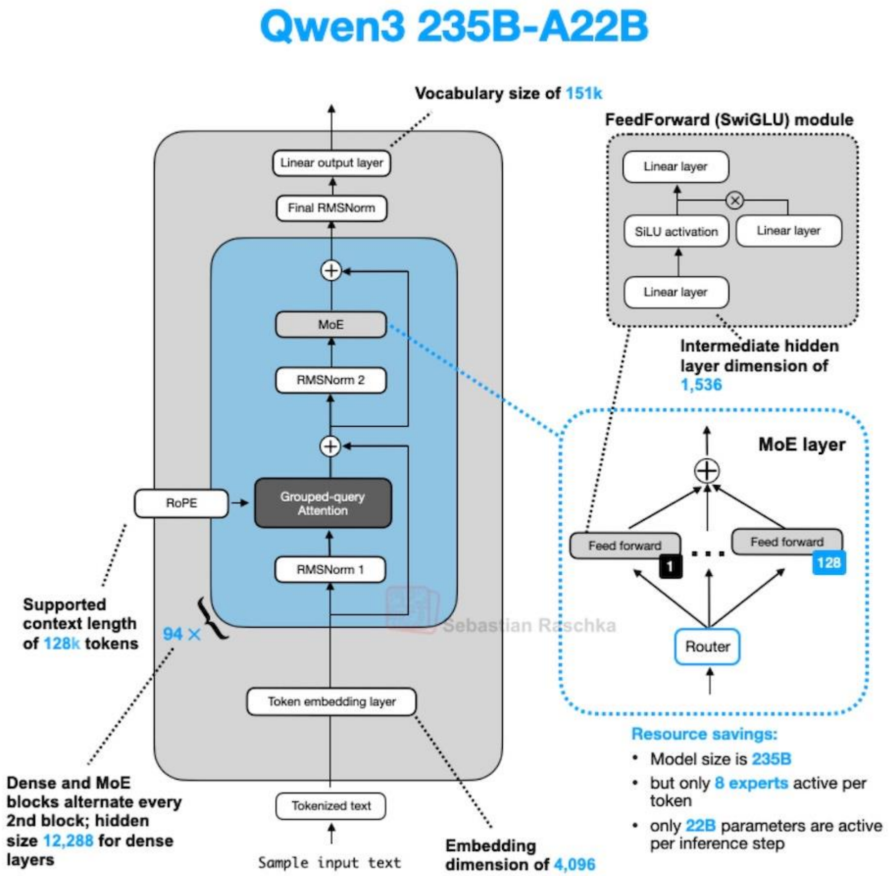
- Efficient linear algebra goes beyond JL:
  - There are non-JL sketching matrices that yield AMM and OSE
  - They lead to even faster solutions for these problems
  - We might see a bit of this in HW3

Questions?

Next: Applications of LSH and KDE to LLMs

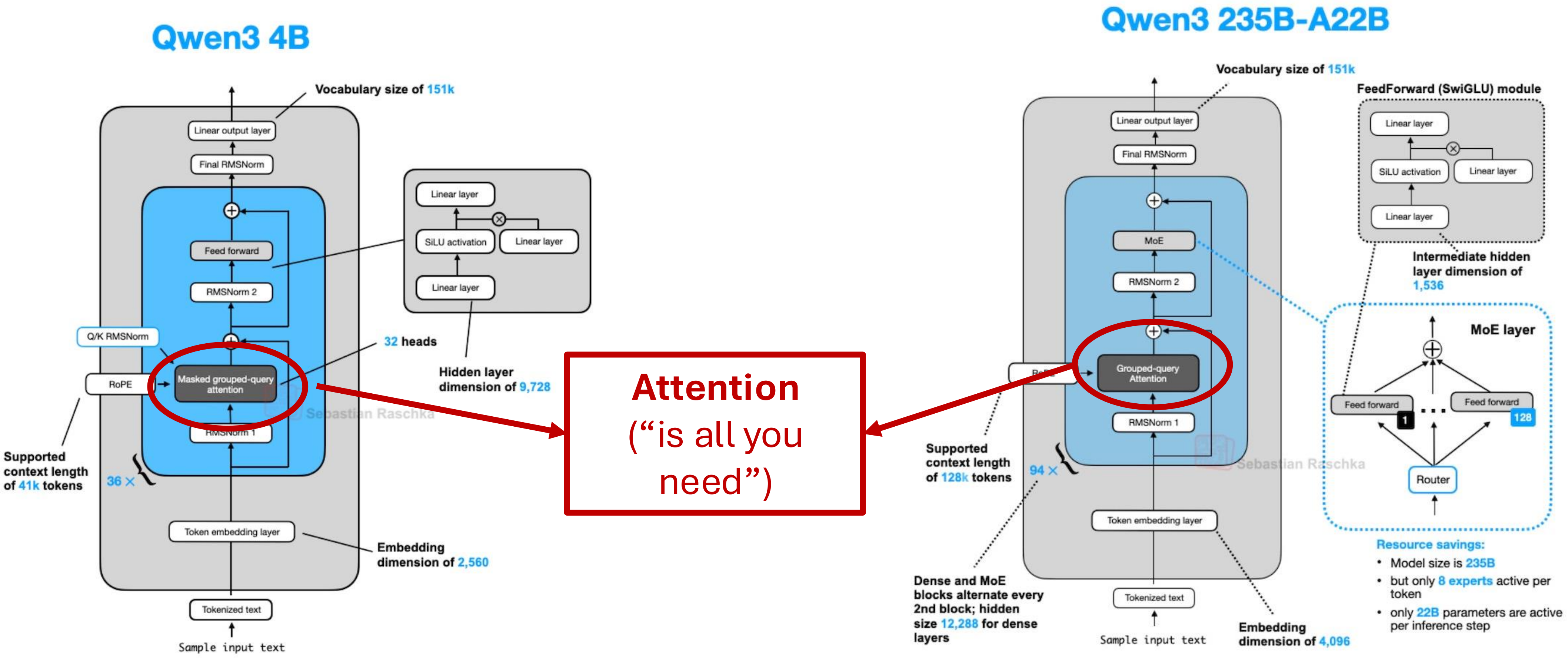
Crash Course on Large Language Models





[Credit: Sebastian Raschka]()

Crash Course on Large Language Models



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Attention

Attention

- **Definition – Softmax:**
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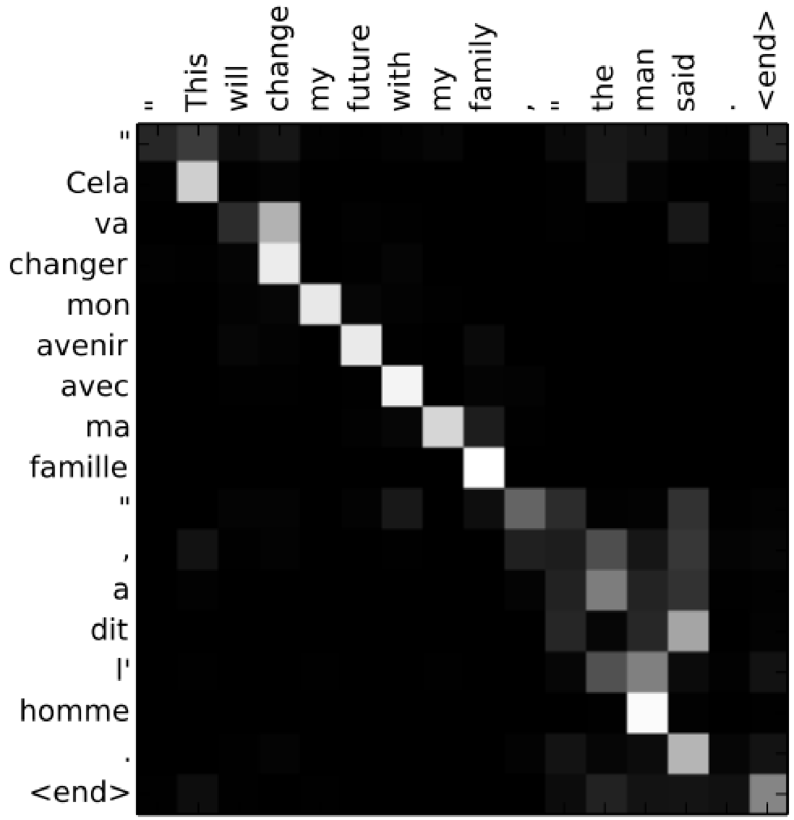
  - n is the **"context length"** – number of tokens the LLM needs to process at once – **major bottleneck in AI** – **usually requires efficient approximations**

Attention - Intuition

- The goal of the Attention matrix A is to describe which tokens are related to which tokens in the sequence



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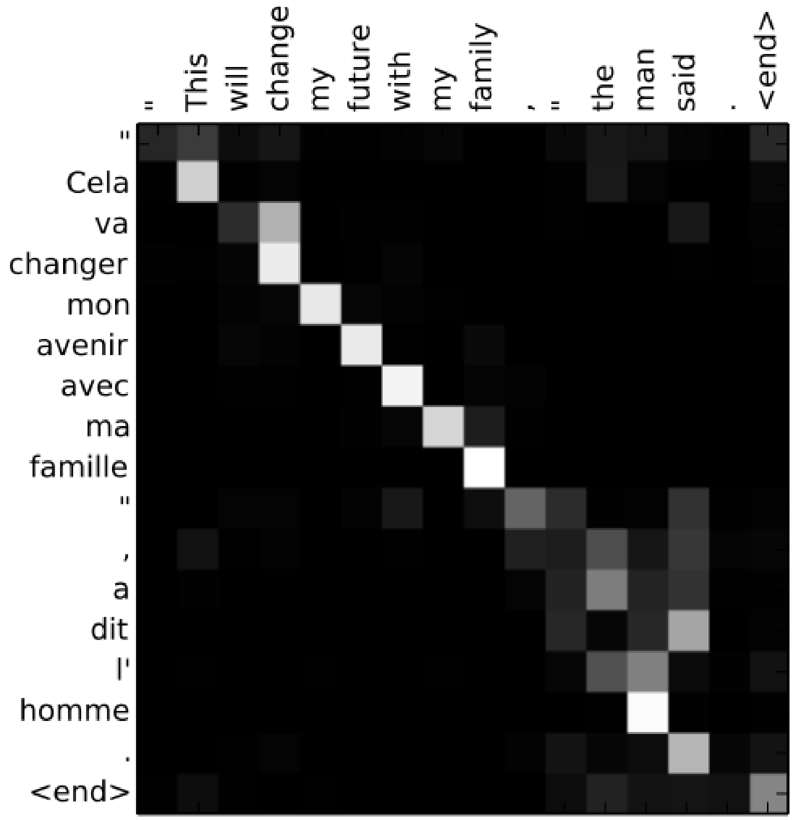

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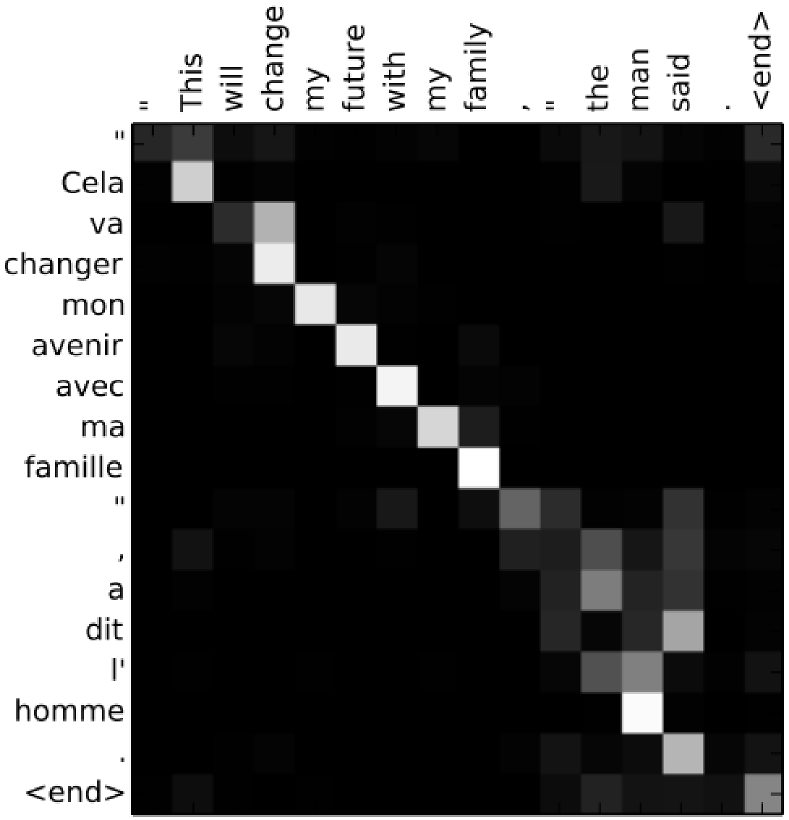

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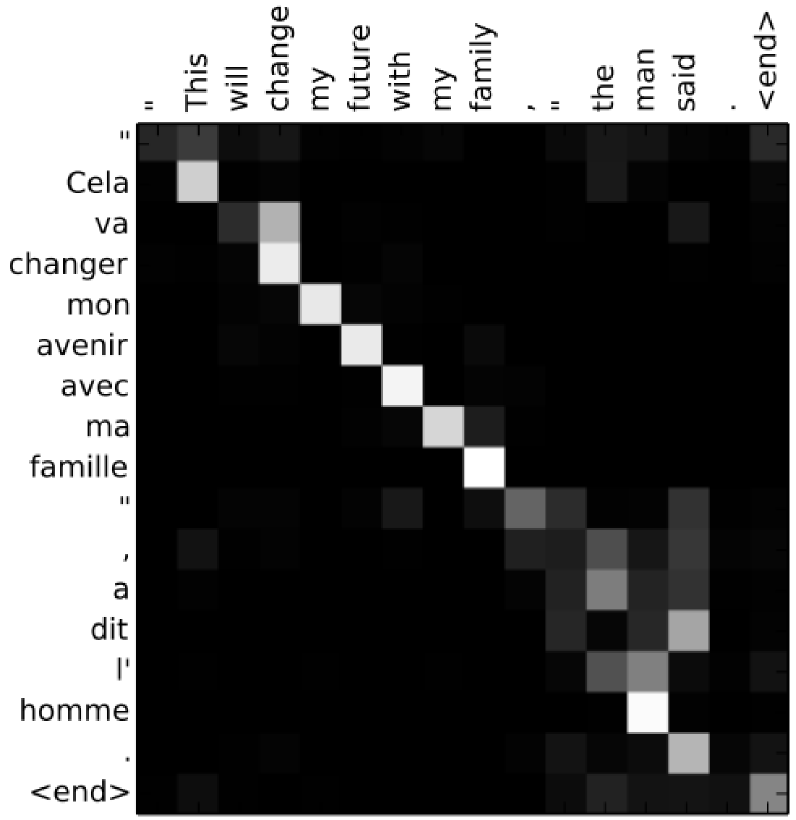

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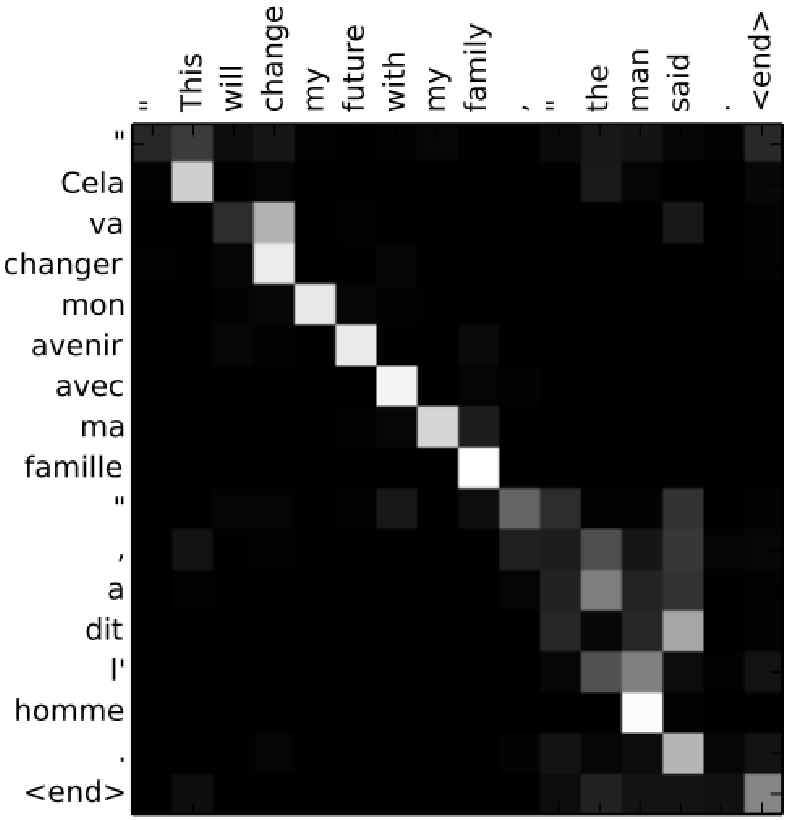
  - The final embeddings are used for next-token prediction



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- **Upshot:**
  - We have two sets of vectors: queries $q_1, \dots, q_n$ and keys $k_1, \dots, k_n$ in $\mathbb{R}^d$
  - We have a "softmax kernel": $k(x, y) \propto e^{q_i^T k_j}$
  - The Attention matrix A is the corresponding kernel matrix
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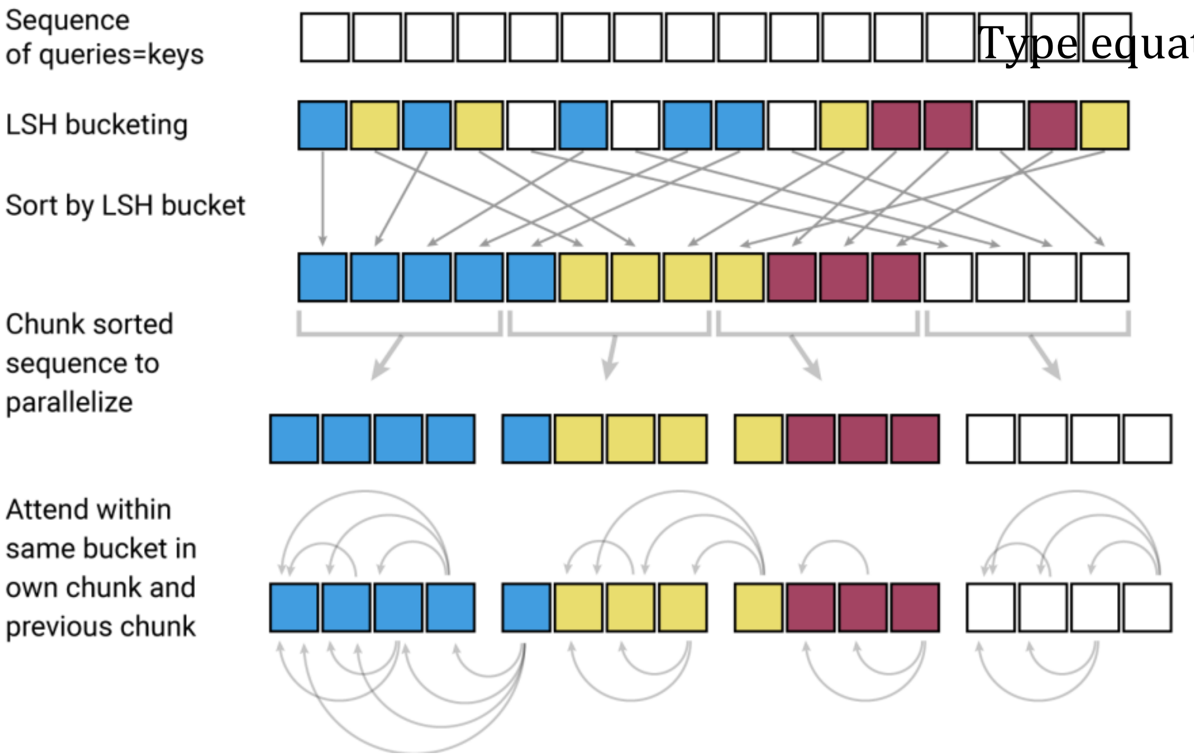
- Have we seen techniques that could be relevant?

- There are very many papers taking this approach; we'll see a small sample
  - Disclosure: this is very popular in research papers but less so in practice (why?)

"Reformer": Attention via LSH

[Kitaev, Kaiser, Levskaya; ICLR 2020]

- We want to find query-key pairs with large inner product
- Use LSH!



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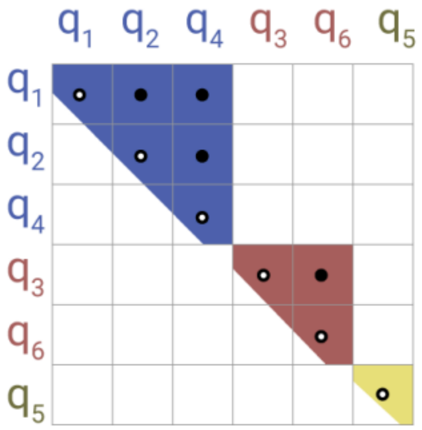
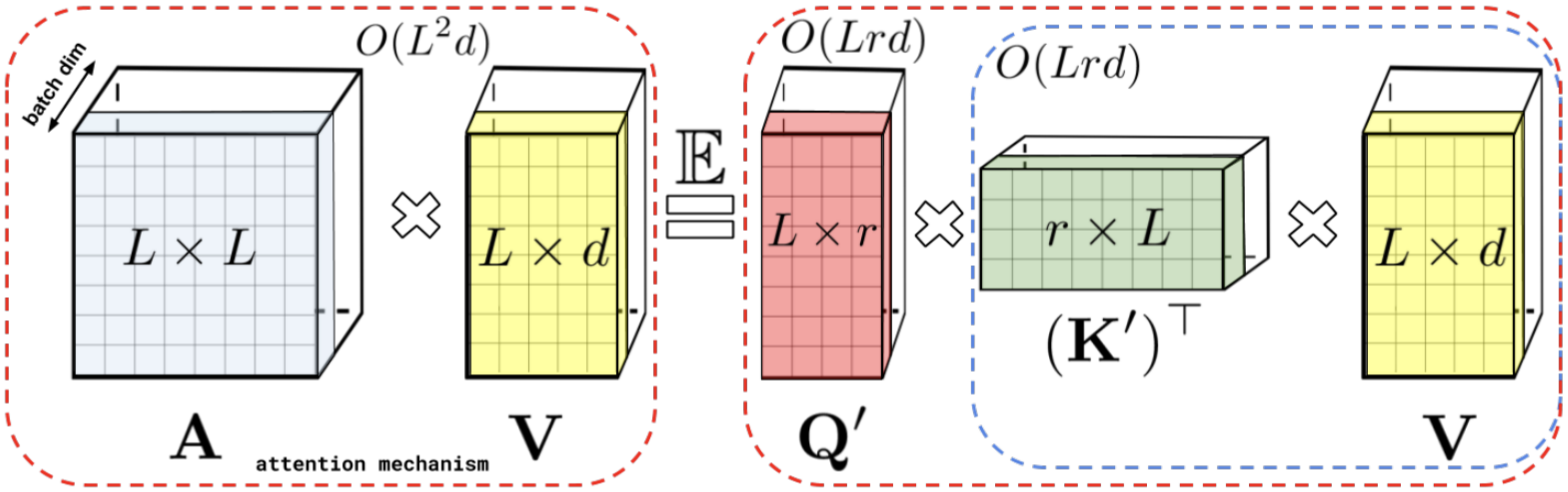


“Performer”: Attention via Random Features

[Choromanski et al.; ICLR 2021]

- Attention is a kernel matrix
- Approximate it with random features!
- They develop *positive* random features because positive/negative random features like we’ve seen (e.g., RFF) lead to too large errors for near-zero values



KDEformer: Attention via Fast KDE

[Zanideh, Han, Daliri, Karbasi; ICML 2023]

- Attention is kernel matrix-vector multiplication
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HyperAttention: Attention via AMM

[Han, Jayaram, Karbasi, Mirrokni, Woodruff, Zanideh; ICLR 2024]

- Attention is matrix-vector multiplication
- Approximate it with sketch-and-solve approximate matrix multiplication!

Thinformer: Attention via KDE Coresets

[Carrell, Gong, Shetty, Dwivedi, Mackey; ICML 2025]

- Attention is a kernel matrix
- Approximate by sampling a small KDE-preserving subset (i.e., coreset!)
- Based on a recently published new “kernel thinning” technique for kernel coresets

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RACE Attention: Attention via LSH Features

[Joshi, Chowdhury, Kanakamedala, Singh, Tu, Shrivastava; ICLR 2026]

- Attention is a kernel matrix
- Approximate it with LSH-based random features!
- The ”softmax kernel” is not LSHable; they replace Attention with an LSHable analog

Concluding Remarks

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    - KV-cache compression by JL (tl;dr: project keys and values to lower dimension and round them to bits)
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- **Questions?**